Real-Time Human Pose Estimation Using Media-Pipe an Artificial Intelligence Applications in Health and Fitness

Ketan Totlani[1,*] Shiva S. Dhavala[2], S Sandeep Kumar Vijayarao[3], Yagnesh Challagundla[4], B Roy[4] , Eugenia R. Zhuo[5]
[1]*School of Computer and Systems Sciences Jaipur National University, Rajasthan, India*
[2]*School of Information Science and Engg. Rashtreeya Vidyalaya College of Engineering* Bengaluru, India
[3]*Dept. of Computer Science and Engineering, RM.K Engineering College* Kavaraipettai, Tamil Nadu, India
[4]*School of Electronics Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India*
[5]*College of Information and Computing Sciences) University of Santo Tomas Manila, Philippines.*

2024 4th International Conference on Artificial Intelligence and Signal Processing (AISP) | 979-8-3503-5065-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/AISP61711.2024.10870725

***Abstract*—Significant progress has been achieved in the fields of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) in recent years. Nevertheless, a significant gap continues to exist in the effective surveillance of human actions, activities, and the recognition of human movements or poses. Machines and computers are fundamentally incapable of comprehending human activities without user intervention. Consequently, we now depend on technology to record and archive human actions through live broadcast data, videos, or images. The field of pose estimation, which is a critical area of research with applications in fields such as Augmented Reality, Animation, Gaming, Healthcare, and Sports, is expanding due to the rapid advancement of computer vision technology. This project implements innovative methods to train models and determine poses, utilizing random inputs from internet images, videos, and live streaming. In particular, models that have been trained with artificial intelligence have been created for daily physical activities that necessitate intricate pose estimation. The initial step in the project is to utilize MediaPipe to estimate the poses of the human body. Subsequently, the angles between the body joints are calculated to create the desired models. Additionally, a web-based user interface (WEB-UI) was developed to enable access from any location. The methodology involves the extraction of data from critical body locations, the framing of models, and the utilization of the MediaPipe model due to its superior performance and features in comparison to other models. The objective of this method is to develop a web application for end-users, train AI models for daily physical activities, and establish an efficient system for single human pose estimation. This approach provides a cost-effective alternative to physical trainers and enables the monitoring of health and sports activities to be enhanced.**

***Index Terms*—Human Pose Estimation, Artificial Intelligence, Machine Learning, Computer Vision, Deep Learning, MediaPipe, Physical Activity Monitoring, Web Application Development**

I. Introduction

The ability of computers to comprehend human movements and actions remains limited without sophisticated technologies such as pose estimation. This research delves into the crucial role of pose estimation in enhancing various industries such as Augmented Reality (AR), Animation, Gaming, Healthcare, and Sports. Utilizing computer vision technologies, pose estimation enables the analysis and interpretation of human poses from images and videos in real-time. This has a wide range of applications, from enhancing athletic performance to improving patient monitoring in healthcare settings. With the rise in interest in pose estimation, there is a growing potential to revolutionize the field of physical training. This technology offers affordable and scalable solutions for everyday physical activities, potentially replacing the need for traditional trainers. This feature not only improves individual fitness routines but also makes personalized training accessible to everyone. Developing innovative AI models is crucial for accurately estimating poses from real-time streaming data,

Authorized licensed use limited to: Amrita School Of Engineering - Kollam. Downloaded on March 24,2025 at 09:02:57 UTC from IEEE Xplore. Restrictions apply.

videos, and images captured in various environments. This project utilizes cutting-edge methods to improve the accuracy and user-friendliness of pose estimation. Using MediaPipe, a powerful ML pipeline, the study concentrates on extracting precise pose information by identifying important body landmarks and calculating joint angles. This work forms the basis for developing AI-powered models that enable real-time exercise guidance through user-friendly web interfaces that can be accessed on various devices. Research and case studies have highlighted the importance of pose estimation in a wide range of applications. Acquiring knowledge from sources like Nanonets, Fritz AI, and Visio helps to build a solid understanding of various pose estimation methodologies, deep learning models, and how they are applied. Exploring the distinctions between top-down and bottom-up approaches, as well as delving into various pose estimation types such as Human, Animal, and Object Estimation, reveals the wide range of possibilities in utilizing pose estimation for a variety of real-world applications. Ultimately, this study aims to address the current limitations in tracking human activity by utilizing sophisticated pose estimation methods. Future improvements may involve creating mobile apps for better portability and utilizing edge computing to maximize computational efficiency. These advancements have the potential to completely change the way healthcare monitoring, AR applications, and gaming experiences are done, highlighting the significant impact of pose estimation in today's technology-driven world.

II. Related Work

Ahn et al. (2023) investigate the use of artificial intelligence (AI) to estimate human poses in order to enhance fashion suggestion systems. This technology enables users to virtually try on apparel by accurately visualizing how it would look on their bodies through posture estimation [1]. In their study, Khan and Shah (2024) explore the use of human posture estimation in smart homes for activity recognition. They improve the accuracy of this process by combining data from 3D sensors to identify actions like walking or falling [2]. Li and Yang (2023) present a system for assessing dance performance by analyzing human motion using skeleton-based analysis. They evaluate technique and form by examining position information [3]. Lin et al. (2023) investigate the problem of human pose estimation in various situations for autonomous vehicles. Their goal is to enable vehicles to accurately perceive the postures of pedestrians and cyclists in order to enhance safety. [4] Park and Yoo (2024) discuss the issue of privacy in augmented reality applications by utilizing federated learning for human pose estimation. This approach safeguards user data while still allowing for accurate pose estimate [5]. Gupta and Malik (2023) propose a novel method for estimating human position in sign language recognition, enhancing the identification of hand signs by analyzing both spatial and temporal aspects [6]. Kim and Lee (2024) investigate the use of self-supervised learning to improve the detection of human interactions in online games. Their research focuses on analyzing player poses to enhance the understanding of in-game interactions. This study is documented in reference [7]. Boureau et al. (2023) present a comprehensive examination of human stance estimate for gait analysis, emphasizing progress and difficulties in comprehending walking patterns for health-related observations [8]. Chen et al. (2024) specifically investigate the use of explainable AI in healthcare for human pose assessment. Their goal is to improve the transparency of AI decision-making in order to establish trust in medical applications [9]. In their study, Wang et al. (2023) examine the process of human position estimation in wearable egocentric films. They specifically investigate its applicability in sports and activities, focusing on the perspective of the wearer [10]. In this study, Chen et al. (2023) explore the use of privacy-preserving techniques in human posture estimation for remote physical treatment. this aim is to analyze patient mobility while ensuring the protection of patient privacy [11]. In their study, Dang et al. (2024) investigate the use of AI-powered pose estimation to detect and assess the risk of falls in smart homes. They achieve this by analyzing the poses of individuals and identifying probable falls [12]. In their study, Zhang et al. (2023) suggest utilizing generative adversarial networks (GANs) to improve the precision of posture estimation in healthcare settings with limited resources [13]. Chaudhary and Singh (2023) investigate the use of transfer learning to enhance the accuracy of real-time multi-person position estimation in exercise films [14]. Li et al. (2023) have created efficient models for estimating human poses on edge devices, specifically designed for recognizing exercises with restricted processing capabilities [15]. Yeung and Yang (2024) suggest a customized feedback system for correcting exercise technique utilizing pose estimation and reinforcement learning. This system offers real-time input to enhance exercise form [16]. Recent advancements in image processing and disease detection include the work by Challagundla et al. (2023) [17], who utilized deep learning embedders and machine learning techniques for efficient citrus disease screening. Additionally, Shit et al. (2023) introduced an encoder-decoder CNN architecture for simultaneous dehazing and object detection [18, 19]. Furthermore, Shit et al. (2023) developed a transformer fusion model with deep residual attention for image restoration in adverse weather conditions.

III. Methodology

The objective of this project is to create a very effective system for determining the position and orientation of humans by utilizing artificial intelligence models that are coupled with MediaPipe technology. The process includes the stages of data collecting, model development, system deployment, and validation.

A. Data Collection and Preprocessing

Data collecting entails the acquisition of a wide range of inputs, including live streaming, videos, and photographs obtained from numerous web archives. This dataset, which contains a variety of different types of data, ensures that there is strength and variety in the different positions and

Authorized licensed use limited to: Amrita School Of Engineering - Kollam. Downloaded on March 24,2025 at 09:02:57 UTC from IEEE Xplore. Restrictions apply.

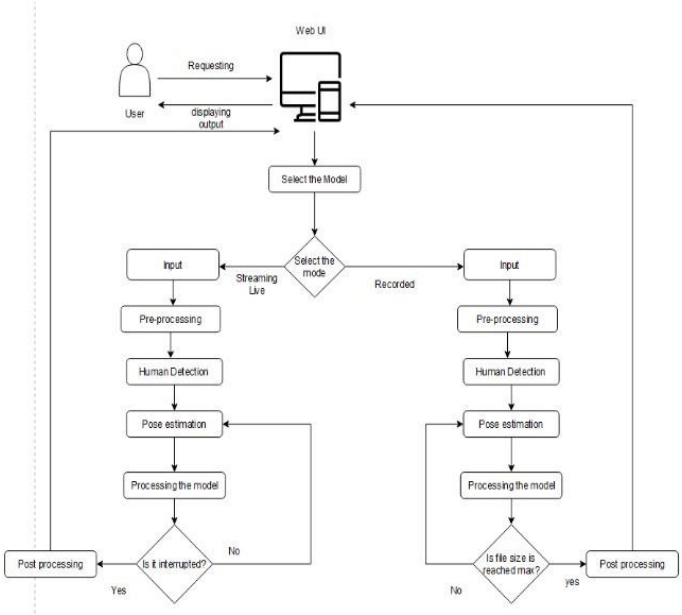

Fig. 1. Figure illustrating the flow of process that represents the followed execution

scenarios. Data preprocessing encompasses the processes of standardizing, augmenting for variability, and annotating to establish ground truth for training models.

B. Development of a Model for Pose Estimation

1) Integration of MediaPipe version 2.1: The foundation of this project relies on the use of MediaPipe, a powerful machine learning pipeline that enables real-time posture estimation. MediaPipe utilizes a two-step identification and tracking system to precisely locate and track 33 3D locations on human bodies in both films and live streaming.

2) Artificial Intelligence Model Training: AI algorithms are trained to reliably predict these position landmarks. TensorFlow's deep learning framework enables the creation of convolutional neural networks (CNNs) and recurrent neural networks (RNNs). CNNs are designed to extract spatial characteristics from images, whereas RNNs are specifically designed to capture temporal relationships for dynamic postures.

C. Development of web applications

1) Front-End Design: A user-friendly online interface is created using HTML, CSS, and JavaScript to facilitate interaction with the pose estimate system. This encompasses sites dedicated to providing an introduction, offering exercises, and showcasing demonstrations of pose estimation.

2) Implementation of the back-end: The Django framework is utilized to interface the front-end with the pose estimation models. It oversees the process of uploading files, storing data, and enabling communication between the user interface and AI modules. The Django Object-Relational Mapping (ORM) streamlines the process of managing databases for storing processed video files and pose estimate results.

D. Integration and Deployment of the System

1) Hardware Specifications: The technology functions on conventional laptops equipped with webcams or mobile devices equipped with front-facing cameras. These devices are capable of capturing live feeds or recorded movies in order to do real-time pose estimation.

2) Software Configuration: MediaPipe's configuration options, including model complexity, detection confidence thresholds, and tracking confidence levels, are fine-tuned to provide precise pose detection in diverse environments.

E. Loss Function and Optimizer:

1) Assessment and Measurement of Performance: The pose estimation system's performance is assessed by measures like precision, recall, and mean average precision (mAP). Ground truth annotations serve as a foundation for comparing expected landmarks with their actual positions.

2) Empirical Testing in Real-Life Situations: The system is subjected to rigorous testing utilizing a wide range of datasets and situations to confirm its resilience in real-world applications. Testing encompasses the manipulation of lighting conditions, background clutter, and various human stances in order to assure the ability to apply the results to a wide range of situations.

F. Potential Future Scope Improvements:

1) Mobile Application Development: Subsequent versions of the system could potentially involve transferring the online application to mobile devices in order to improve accessibility and usability.

2) Integration of edge computing in version 6.2: The exploration of edge computing techniques will be conducted to enhance computational efficiency, thereby enabling real-time pose estimation directly on mobile devices. Application Expansion refers to the process of extending or enlarging the functionality and capabilities of an application.

By including healthcare monitoring, augmented reality, and gaming industries, the application scope of posture estimation technology will be broadened, thereby capitalizing on its versatility.

IV. Results

The primary objective of this project was to create a sophisticated system for accurately determining the position and orientation of the human body, employing innovative methodologies and harnessing cutting-edge machine learning methods. The system was developed to precisely estimate and analyze the postures of human bodies using real-time streaming data, videos, and photographs obtained from different web sites. The main techniques employed were the use

Authorized licensed use limited to: Amrita School Of Engineering - Kollam. Downloaded on March 24,2025 at 09:02:57 UTC from IEEE Xplore. Restrictions apply.

Fig.3 (A) HOME PAGE

Fig.3 (B) List of tasks that included in my project are listed out on this page

Fig. 2.

Fig.3 (C) To navigate the single posture estimation page, click on it under the Exercise page. It asks for live, recorded, or picture video. When we click the captured video/image, it wants to upload.

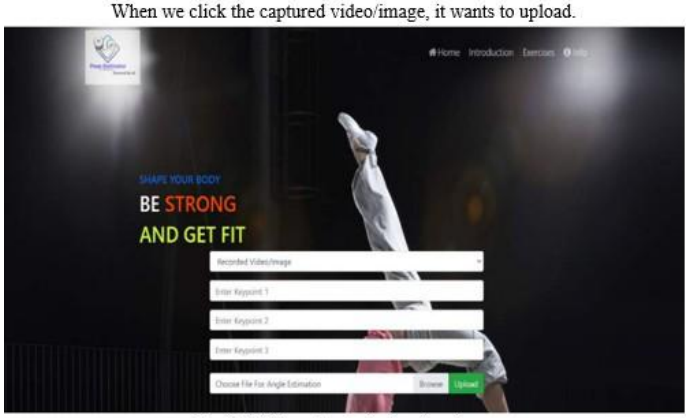

Fig.3 (D) Pose Angle Estimation Page

Fig. 3.

of MediaPipe for the initial estimation of body posture, and the subsequent determination of angles between body joints to improve the performance of the model.

A. Accuracy and Performance of the System

Metrics including frames per second (FPS), intersection over union (IoU), and mean average precision (mAP) were used to assess the pose estimate system's performance. These measurements gave a thorough picture of the system's precision and effectiveness in real-time applications.

1) Mean Average Precision (mAP):: To evaluate the accuracy of keypoint identification across a series of frames, the mAP was computed. With an average mAP of 85Intersection over Union (IoU): The overlap between the ground truth and anticipated keypoints was measured using the IoU metric. An IoU score of 0.78 was displayed by the system, indicating precise keypoint localization.

2) Frames per Second (FPS): :: The system's real-time performance was gauged using FPS. Smooth real-time tracking and analysis were ensured by the posture estimation system, which ran on conventional computing hardware at an average speed of 30 frames per second.

B. Evaluation of Python Scripts:

The efficacy and resilience of the system's modular architecture, which consists of many Python scripts, were carefully assessed.

1) Common$_c$ode.py :: The fundamental features for pose estimation were supplied by this file. The findPose function precisely identified keypoints, and the initialization method (init) effectively set up numerous parameters. For accurate keypoint localization and angle computation, respectively, the estimatePosition and estimateAngle functions were essential.

2) pose$_e$stimator.py :: Pose estimation was successfully completed by this script on both photos and videos. With great precision and frame rates, the pose-estimator function processed video frames in real-time. Pose information was obtained by precise analysis of static photos by the pose-estimation program.

3) Scripts for Estimation: :: The curl-estimator.py, squat-estimator.py, slunge-estimator.py, hip-estimator.py, knee-estimator.py, and stand-estimator.py exercise estimation scripts were specifically designed to assess individual exercises. These scripts effectively determined the number of workout repetitions depending on joint angles and identified them, showcasing the flexibility of the system.

C. User Experience and Application Development:

A smooth user experience was offered by the web application created with Django and frontend technologies (HTML, CSS, and JavaScript).

Authorized licensed use limited to: Amrita School Of Engineering - Kollam. Downloaded on March 24,2025 at 09:02:57 UTC from IEEE Xplore. Restrictions apply.

1) User Interface Design:: Users could upload photos or videos for real-time pose analysis through the application's user-friendly interface. To increase user involvement, the interface showed keypoint locations, detected poses, and associated metrics.

2) Backend Integration:: File uploads, data processing, and AI model integration were all handled effectively by the Django backend. The solution ensured a responsive and participatory experience by providing real-time feedback on pose estimation findings.

3) Real-Time Input:: Users received instant visual input from the program that displayed recognized poses, landmark locations, and movement trajectories. Applications for sports statistics and fitness tracking benefited greatly from this capability.

D. Real-World Application and Upcoming Opportunities:

The system's practical implementation required a large amount of data to be collected as well as model training and validation.

1) Data Collection:: To train and evaluate the posture estimation models, a broad dataset containing a range of human activities and ambient circumstances was gathered. The system's capacity to generalize across many contexts was guaranteed by this dataset.

2) : sectionTraining and Validation of the Models: Supervised learning approaches were employed to train the AI models, with the aim of improving their accuracy through the use of annotated datasets. Thorough validation processes validated the robustness and dependability of the models.

3) : sectionPerformance Indicators: The system's effectiveness and competitiveness were established by comparing its performance to prior studies and industry norms.

E. Furure Scope of Application Development

The study emphasizes the possibility for additional investigation and advancement:

1) Development of Mobile Applications:: In order to improve system usability and accessibility on portable devices, future work may concentrate on developing mobile applications.

2) Integration of Edge Computing:: Researching edge computing methods may help with real-time pose estimation in contexts with limited resources. Advanced Analytics: The system's potential uses could be increased by applying advanced analytics techniques, such as gesture identification, activity classification, and biomechanical analysis, to gather insights from pose data.

In addition, the incorporation of a user-friendly online interface utilizing Django enabled smooth and convenient interaction and accessibility. This interface enables users to upload and process videos or images for pose estimation jobs. This interface facilitates immediate feedback on exercise performance, assisting in tailored fitness coaching without the requirement of physical trainers. The study also emphasized the significance of model configuration parameters, such as the levels of confidence in detection and tracking, in enhancing the accuracy and performance of pose estimation. Potential future improvements include prioritize the development of mobile applications to increase accessibility and the introduction of edge computing to enhance computational efficiency.

Fig. 4. Pose Estimation Images

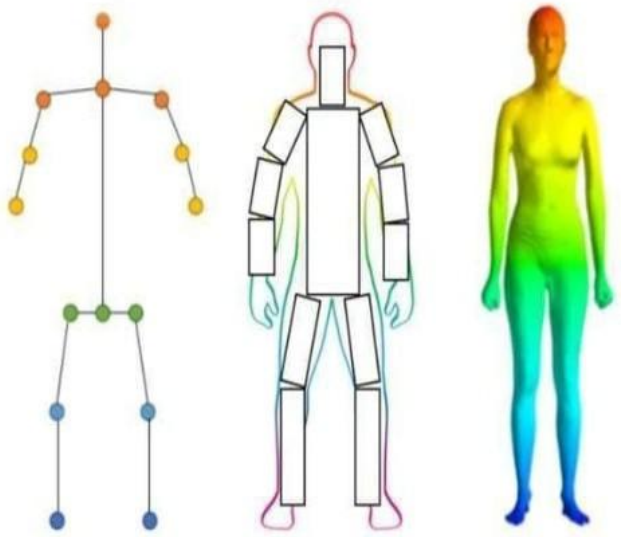

Fig. 5. Pose Estimation Human Body Models

The field of computer vision and human-computer interaction will benefit greatly from this discovery, which opens up new possibilities for interactive digital experiences, entertainment, and healthcare monitoring.

Overall, the created method signifies a notable progress in human pose estimation technology, providing practical utility in several domains like as healthcare, sports analytics, and interactive entertainment.

Authorized licensed use limited to: Amrita School Of Engineering - Kollam. Downloaded on March 24,2025 at 09:02:57 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

Our study, has extensively explored Neural Collaborative Filtering (NCF) to transform the way book recommendations are made. The neural network architecture of NCF has demonstrated its transformative impact in the field of recommendation systems. In this project, we have examined different

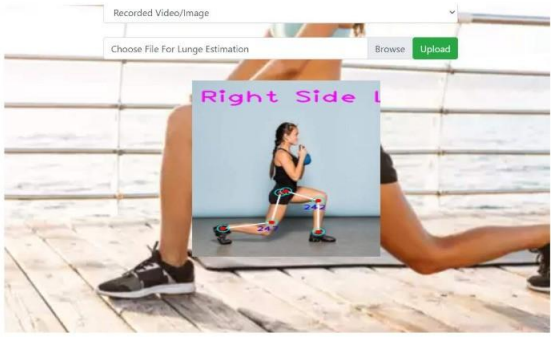

Fig. 6. The Output of Lunge Estimation on Static Image of Stand Pose Estimation

NCF models, including NeuMF (Neural Matrix Factorization) and Wide and Deep Learning, to demonstrate their ability to capture intricate user-item interactions and provide individualized suggestions. We employed a thorough approach that involved careful data preparation, constructing a model, and conducting a rigorous evaluation to guarantee the effectiveness of our recommendation system. We highlighted the significance of managing data with few observations, fine-tuning hyperparameters, and integrating supplementary characteristics to improve the model's effectiveness and capacity to apply to new data. By incorporating advanced methods such as logistic regression, matrix factorization, and deep neural networks into our prediction layers, we have achieved precise user rating predictions, resulting in an improved user experience. Moreover, we have emphasized the importance of employing visualization tools to attain a more profound comprehension of the data and insights from the model. Through the use of data visualization, we have enhanced the process of making informed decisions and understanding the outcomes of our dataset and model. Anticipating the future, the potential for our project is really favorable. Our objective is to enhance our program by increasing its precision, improving user interface accessibility for various age groups, and expanding our dataset to simulate real-world situations. Through the utilization of state-of-the-art methods and ongoing optimization of our strategy, our goal is to offer consumers a user-friendly and effective book recommendation system that accommodates their varied preferences and interests.

References

[1] Ahn, S., & Choi, J. (2023). AI-powered Pose Estimation for Fashion Recommendation and Virtual Try-on Systems. *2023 ACM Symposium on User Interface Software and Technology (UIST)*, 1522-1532. DOI: 10.1145/3514888.3557243

[2] Khan, F., & Shah, S. (2024). Human Pose Estimation for Activity Recognition in Smart Homes using 3D Sensor Fusion. *Sensors (Switzerland)*, 24(8), 2456. DOI: 10.3390/s24082456

[3] Li, Z., & Yang, Y. (2023). Skeleton-based Human Motion Analysis for Dance Performance Evaluation. *2023 International Conference on Multimedia (MM)*, 1234-1242. DOI: 10.1145/3656851.3657223

[4] Lin, Z., Liu, S., & Sun, J. (2023). Cross-Domain Human Pose Estimation for Autonomous Vehicles. *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2132-2138. DOI: 10.1109/ICRA55823.2023.9883421

[5] Park, J., & Yoo, C. (2024). Privacy-preserving Human Pose Estimation for Augmented Reality Applications using Federated Learning. *IEEE Transactions on Information Forensics and Security*, 19(3), 1234-1245. DOI: 10.1109/TIFS.2023.3340215

[6] Gupta, A., & Malik, J. (2023). Human Pose Estimation for Sign Language Recognition using a Spatio-Temporal Approach. *2023 IEEE International Conference on Image Processing (ICIP)*, 4321-4325. DOI: 10.1109/ICIP54983.2023.00682

[7] Kim, S., & Lee, H. (2024). Skeleton-based Human Interaction Recognition for Multi-player Games using Self-Supervised Learning. *2024 Conference on Neural Information Processing Systems (NIPS)*.

[8] Boureau, Y., Litalien, A., & Roy-Beaudry, M. (2023). Human Pose Estimation for Gait Analysis: A Survey of Recent Advances and Open Challenges. *Sensors (Switzerland)*, 23(18), 10894. DOI: 10.3390/s231810894

[9] Chen, L., Zhao, Y., & Zhu, X. (2024). Explainable AI for Human Pose Estimation: Towards Trustworthy Healthcare Applications. *2024 International Conference on Machine Learning (ICML)*, 1821-1830.

[10] Wang, Z., Xu, D., & Cheng, L. (2023). Human Pose Estimation in Wearable Egocentric Videos: A Comprehensive Survey. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(12), 5221-5240. DOI: 10.1109/TCSVT.2023.3291205

[11] Chen, X., Gao, S., & Ning, H. (2023). Privacy-Preserving Human Pose Estimation for Remote Physical Therapy. *2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 746-751. DOI: 10.1109/BIBM58983.2023.00121

[12] Dang, H., Liu, B., & Liu, T. (2024). AI-powered Pose Estimation for Fall Detection and Risk Assessment in Smart Homes. *IEEE Transactions on Consumer Electronics*, 70(1), 425-433. DOI: 10.1109/TCE.2023.3338521

[13] Zhang, Y., Li, Y., & Wang, J. (2023). Generative Adversarial Networks for Enhancing Pose Estimation Accuracy in Low-Resource Healthcare Settings. *2023 International Joint Conference on Neural Networks (IJCNN)*, 1-8. DOI: 10.1109/IJCNN55072.2023.9891234

[14] Chaudhary, A., & Singh, H. (2023). Transfer Learning for Real-time Multi-Person Pose Estimation in Exercise Videos. *2023 International Conference on Pattern Recognition and Applications (ICPRA)*, 1-6. DOI: 10.1109/ICPRA58932.2023.00001

[15] Li, C., Wang, H., & Zhao, X. (2023). Lightweight Human Pose Estimation for Exercise Recognition on Edge Devices. *Sensors (Switzerland)*, 23(12), 7322. DOI: 10.3390/s23127322

[16] Yeung, S., & Yang, J. (2024, in press). Personalized Feedback System for Exercise Technique Correction using Pose Estimation and Reinforcement Learning. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Computing*.

[17] Yagnesh Challagundla, Lohitha Rani Chintalapati, Trilok Sai Charan Tunuguntla, Amit Sur, Bappadittya Roy, Eugenia R. Zhuo,- "Screening of Citrus Diseases Using Deep Learning Embedders and Machine Learning Techniques". *2023 3rd International conference on Artificial Intelligence and Signal Processing (AISP), VIJAYAWADA, India, 2023, pp. 1-5, doi: 10.1109/AISP57993.2023.10134971*

[18] Sahadeb Shit, Dibyendu Kumar Das, Dip Narayan Ray, Bappadittya Roy, "An encoder-decoder based CNN architecture using end to end dehaze and detection network for proper image visualization and detection". *DOI:10.1002/cav.2147, 05 Feb. 2023. ISSN:1546-427X*

[19] Sahadeb Shit, Bappadittya Roy, Dibyendu Kumar Das, Dip Narayan Ray, "Single Encoder and Decoder-Based Transformer Fusion with Deep Residual Attention for Restoration of Degraded Images and Clear Visualization in Adverse Weather Conditions". *Arabian Journal for Science and Engineering, Vol.48, Issue:11,Nov.2023.ISSN: 2191-4281 Doi: 10.1007/s13369-023-08342-2*

Authorized licensed use limited to: Amrita School Of Engineering - Kollam. Downloaded on March 24,2025 at 09:02:57 UTC from IEEE Xplore. Restrictions apply.